VERILOG Based Simulation of ASK,FSK,PSK,QPSK Digital Modulation Techniques

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***Abstract*—This paper presents a general architectural overview regarding elementary method of VERILOG HDL based code simulation for fundamental and widely used digital modulation techniques such as Binary Amplitude-shift keying (BASK), Binary Frequency-shift keying (BFSK), Binary Phase-shift keying (BPSK) and Quadrature Phase Shift Keying(QPSK). In this work the idea of sinusoidal signals that have been generated is plain sailing in nature and based on fundamentals of signal sampling and quantization. Such concept of sinusoidal signals generation is not unfamiliar but somehow simplified using sampling and quantization in time and amplitude domain, respectively. The whole simulation is done on Model Sim and Xilinx-ISE using VERILOG Hardware descriptive language. The work has been accomplished on Thirty two bit serial data transmission with self-adjustable carrier frequency and bit duration length.**

Keywords- Orthonormal, Analog, FPGA, BASK, BFSK, BPSK, QPSK, OQPSK, QAM, MSK, CPFSK, PISO

I. INTRODUCTION

Over the previous few decades, there has been occurred a big alteration from simple analog modulation and angle (phase/frequency) modulation methods to latest digital modulation techniques. Such digital modulation techniques have become fundamentals of satellite communication, Wireless networks and Cellular networks. In the field of telecommunication, modulation is an exercise of diverging either amplitude or angle of a periodic waveform, known as the carrier signal using a modulating signal that in general encompass information to be transmitted, called as intelligence signal. The intelligence signal can be analog or digital in nature that very much depends upon message source. The mechanism of modulation uplifts the frequency band spectrum of transmitted signal to the range, where signals are not much attenuated in the channel. Modulation using a sinusoidal waveform transfigures a baseband electrical signal into a passband signal.

The rudimentary goal of the author is to achieve a general and simplified method to simulate above stated modulation techniques to transmit data serially. The algorithm is proposed in VERILOG HDL and output signals have been simulated on Xilinx-ISE and further analyzed using Model Sim(Wave).

Formation of this paper is: Section 2 provides elementary mathematical analysis regarding BASK, BPSK, BFSK and QPSK digital modulation techniques and representation of modulated signals in terms of orthonormal signals. Section 3 talks about obtained results and conclusions based on simulation outputs. Section 4 and 5 finally end the with Conclusion and acknowledgment

A. *Digital Modulation techniques*

The advancement from conventional analog to digital modulation [1] provides higher information remuneration, more compatibility with digital data assistance, higher data hostage, better superiority in communications, and meteoric system availability [2]. These are the major goals of better communication system establishments that force us to switch from analog to digital domain. In digital modulation, an analog periodic carrier signal, generally sinusoidal is modulated by a discrete level signal [3]. The process of modulation is performed at sender side while the reverse operation i.e. demodulation [4] is done at receiver side.

The basics of all modulation techniques [1] lie in varying three major properties of carrier sinusoidal signal.

$$c(t) = ACos(2\pi * f_c * t + \Phi); \quad (1)$$

where A- Amplitude of carrier signal ;
f_c -Frequency of carrier signal ;
Φ -Phase of carrier signal ;

On the basis of variation of the above three parameters of carrier signals there exists a large number of digital modulation techniques,
ASK, FSK, PSK, QPSK, MSK [5], CPM, CPFSK, QAM etc

[6].

Major modulation techniques are, ASK, FSK, PSK, QPSK.

1) Binary Amplitude shift Keying: Amplitude-shift keying (ASK) [1] is a genre of amplitude modulation that represents digital data as dissimilitude in the amplitude of the carrier signal. In an ASK modulating system, the binary data 1 is symbolized by transmitting a fixed-amplitude carrier wave and fixed frequency for a bit duration of T units. If the signal value is 1 then the carrier signal will be transmitted; otherwise, nothing will be transmitted. Here frequency and phase are kept unchanged throughout signal transmission. It is the most straightforward method and also known as ON-OFF switching.

$$S_1(t) = \frac{2Eb}{Tb} Cos(2\pi f_c t), 0 \le t \le T \tag{2}$$

$$S_2(t) = 0, 0 \le t \le T \tag{3}$$

where Eb is the transmitted signal energy per bit

$$Eb = (t) = A^2 \frac{Tb}{2} \tag{4}$$

$$A = \frac{2Eb}{Tb} \tag{5}$$

There is a single basis function $\Phi_1(t)$ and is given by:

$$\Phi_1(t) = \frac{2}{Tb} Cos(2\pi f_c t), 0 \le t \le T \tag{6}$$

$$S_1(t) = \sqrt{Eb}\Phi_1(t), 0 \le t \le T \tag{7}$$

$$S_2(t) = 0, 0 \le t \le T \tag{8}$$

Constellation diagram for ASK is shown in Fig. 1.

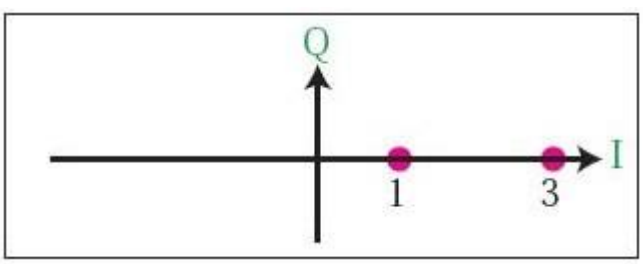

Fig. 1. Constellation diagram of ASK

2) Binary Frequency shift Keying: Frequency-shift keying (FSK) [3] is analogous to fundamental frequency modulation technique in which digital information is transmitted through deviating the frequency of the carrier signal. The carrier frequency is shifted in according to the input data stream, phase and amplitude of the carrier are maintained constant. In binary FSK (BFSK) two different carrier frequencies ($f_1 and f_2$) [1] are used and symbols 1 and 0 are represented by $S_1(t) and S_2(t)$ are defined as

$$S_1(t) = \frac{2Eb}{Tb} Cos(2\pi * f_1 * t), 0 \le t \le T \tag{9}$$

$$S_2(t) = \frac{2Eb}{Tb} Cos(2\pi * f_2 * t), 0 \le t \le T \tag{10}$$

There is set of orthonormal basis [1] function $_1(t) and _2(t)$ and is given by:

$$\Phi_1(t) = \frac{2}{Tb} * Cos(2\pi * f_1 * t), 0 \le t \le T \tag{11}$$

$$\Phi_2(t) = \frac{2}{Tb} * Cos(2\pi * f_2 * t), 0 \le t \le T \tag{12}$$

$$S_1(t) = \sqrt{Eb} * \Phi_1(t), 0 \le t \le T \tag{13}$$

$$S_2(t) = \sqrt{Eb} * \Phi_2(t), 0 \le t \le T \tag{14}$$

3) Binary phase shift Keying: The general Phase-shift keying (PSK) [4] is such kind of modulation scheme that transmit data via changing(modulating) the phase of a carrier signal, while keeping amplitude and frequency constant during data transmission. In binary PSK(BPSK) two carrier signals are used for transmission which have phase difference of 180.

$$S_1(t) = \frac{2Eb}{Tb} Cos(2\pi * f * t), 0 \le t \le T \tag{15}$$

$$S_2(t) = \frac{2Eb}{Tb} Cos(2\pi * f * t + \pi), 0 \le t \le T \tag{16}$$

There is only one orthonormal function [1] would be enough in BPSK.

$$\Phi(t) = \frac{2}{Tb} * Cos(2\pi * f * t), 0 \le t \le T \tag{17}$$

Both the transmitted signal can be represented in terms of the orthonormal signal

$$S_1(t) = \sqrt{Eb} * \Phi(t), 0 \le t \le T \tag{18}$$

$$S_2(t) = -\sqrt{Eb} * \Phi(t), 0 \le t \le T \tag{19}$$

Constellation Diagram for PSK is shown below: 2

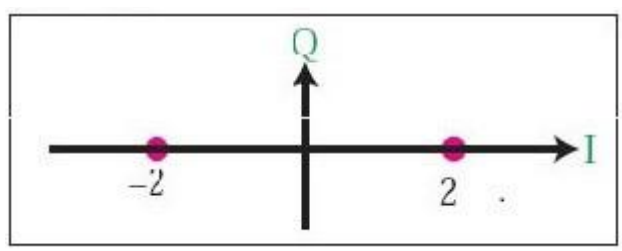

Fig. 2. Constellation diagram of PSK

4) Quadrature phase shift Keying: Quadrature phase shift keying(QPSK) technique contains more efficiency in itself compared to ASK, PSK and FSK in terms of data transmission rate and bandwidth utilization and is also without affecting bit error rate at the cost of more complex receiver structure [7]. It is also defined as composition of frequency shift keying and phase shift keying. In QPSK two carriers of same frequency which are orthogonal to each other are used for transmission of two bit at a time with equal bandwidth and bit error rate to PSK [1].

In QPSK two bits are transmitted at a time, so there can be four possible symbols 00,01,10 and 11.

$$S_1(t) = \frac{2Eb}{Tb} Cos(2\pi * f * t + \frac{\pi}{4}), 0 \le t \le T \quad (20)$$

$$S_2(t) = \frac{2Eb}{Tb} Cos(2\pi_* f_* t + \frac{3*\pi}{4}), 0 \le t \le T \quad (21)$$

$$S_3(t) = \frac{2Eb}{Tb} Cos(2\pi_* f_* t + \frac{5*\pi}{4}), 0 \le t \le T \quad (22)$$

$$S_4(t) = \frac{2Eb}{Tb} Cos(2\pi_* f_* t + \frac{7*\pi}{4}), 0 \le t \le T \quad (23)$$

There is set of orthonormal basis function $_1(t) and _2(t)$ and is given by:

$$\Phi_1(t) = \frac{2}{Tb} * Cos(2\pi * f_1 * t), 0 \le t \le T \quad (24)$$

$$\Phi_2(t) = \frac{2}{Tb} * Cos(2\pi * f_2 * t), 0 \le t \le T \quad (25)$$

All the four symbols are represented in orthonormal signals $_1(t) and _2(t)$ as-

$$S_1(t) = \frac{2Eb}{Tb} Cos(2\pi * f * t + \frac{\pi}{4}), 0 \le t \le T \quad (26)$$

$$S_2(t) = \frac{2Eb}{Tb} Cos(2\pi_* f_* t + \frac{3*\pi}{4}), 0 \le t \le T \quad (27)$$

$$S_3(t) = \frac{2Eb}{Tb} Cos(2\pi_* f_* t + \frac{5*\pi}{4}), 0 \le t \le T \quad (28)$$

$$S_4(t) = \frac{2Eb}{Tb} Cos(2\pi_* f_* t + \frac{7*\pi}{4}), 0 \le t \le T \quad (29)$$

Constellation diagram for QPSK is shown in below figure: 3

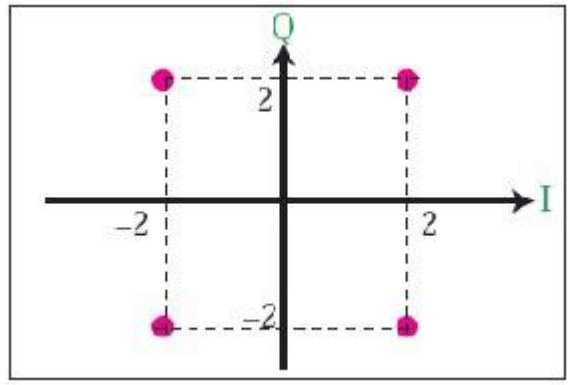

Fig. 3. Constellation diagram of QPSK

TABLE I
POWER SPEED AND AREA COMPARISION AMONG ALL MODULATION TECHNIQUES

Parameters	Values			
	BASK	BFSK	BPSK	QPSK
Power	0.039 W	0.040 W	0.040 W	0.042 W
Number of Occupied Slices	210	210	72	44
Number of 4-input LUTs	147	147	56	25
Number of bonded IOBs	90	90	60	9
Rise time of clock	9.309 ns	10.260 ns	9.801 ns	9.683 ns

II. RESULTS AND DISCUSSIONS

A. ARCHITECTURAL OVERVIEW

All the simulated modulation techniques take 32 bits inputs as the parallel data input which is need to be transmitted and one global system clock which runs every component present in system. The ground level architecture of all modulation techniques are illustrated through block diagram in Fig.4. The primary component of the architecture, the sinusoidal signal generator has been simulated by deviating the register value continuously in periodic manner of deviation. Cosine and inverted sinusoidal signal can be made by adjusting the deviation and varying time period. At the initial stage a parallel input serial output shift register [8] is employed to convert the parallel data bits into serial stream Fig.4. Here the clock retained to the piso is slowed down using a Mod-32(5 bit) ripple carry counter [**?**] for achieving desired bit duration length. So variable bit length can be achieved from the method. For an example if global system clock of 50 MHz(For SPARTAN 3AN) [9] and the frequency divider counter is of Mod-32 than the bit duration length will be about0.64 microseconds.

The further framework of architecture depends upon the particular modulation techniques. The simulated waveforms of above stated modulation techniques are shown . The major trade off among power, speed and area [10] for all four techniques are shown in Table. II-A.

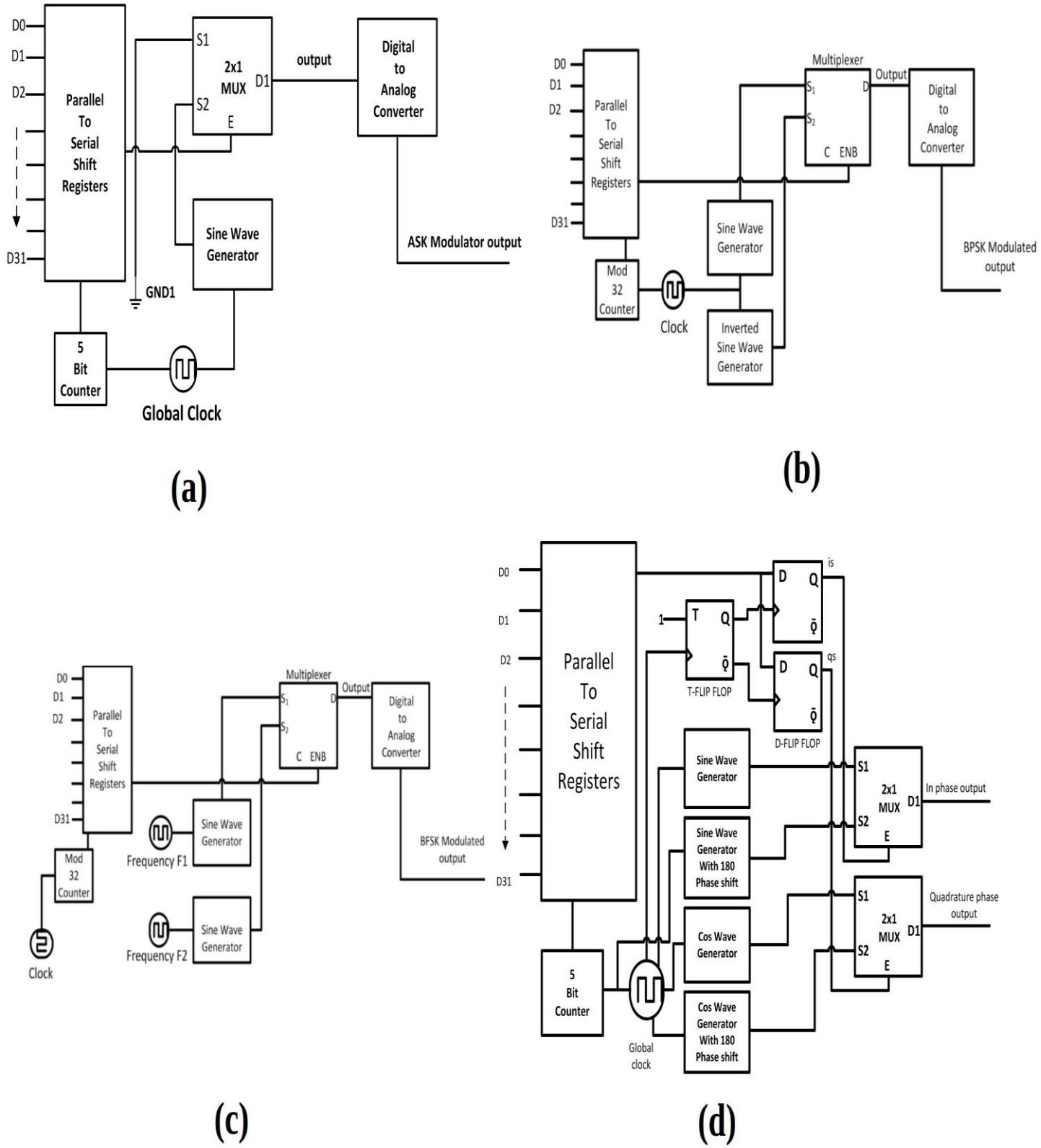

Fig. 4. (a) Block Diagram for BASK Implementation (b) Block Diagram for BPSK Implementation (c) Block Diagram for BFSK Implementation (d) Block Diagram for QPSK Implementation

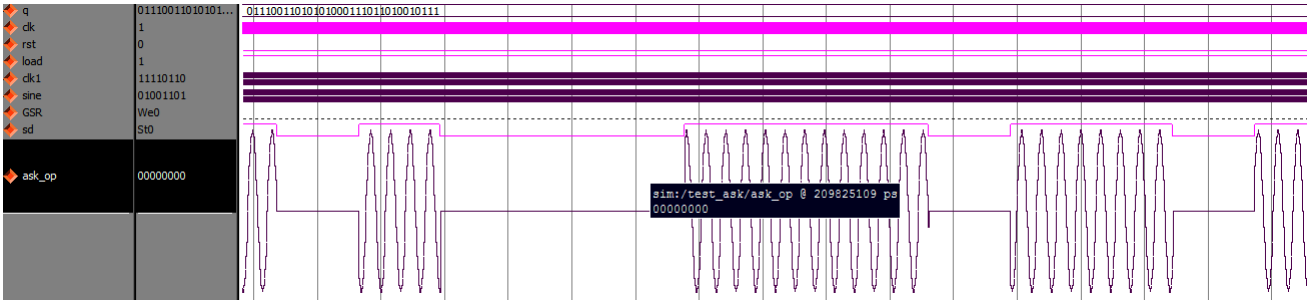

Fig. 5. Simulated output of BASK

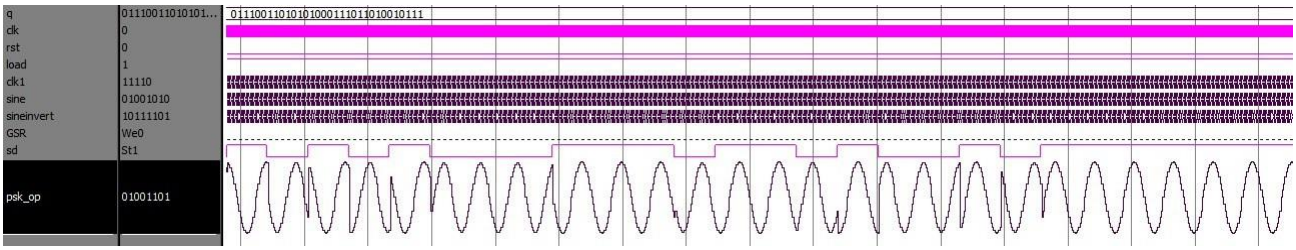

Fig. 6. Simulated output of BPSK

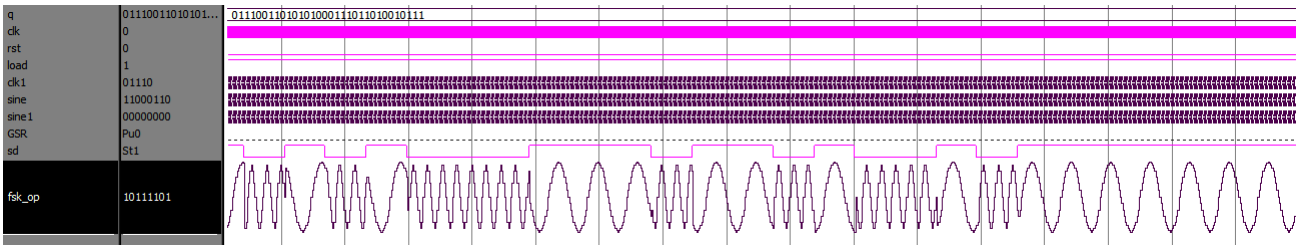

Fig. 7. Simulated output of BFSK

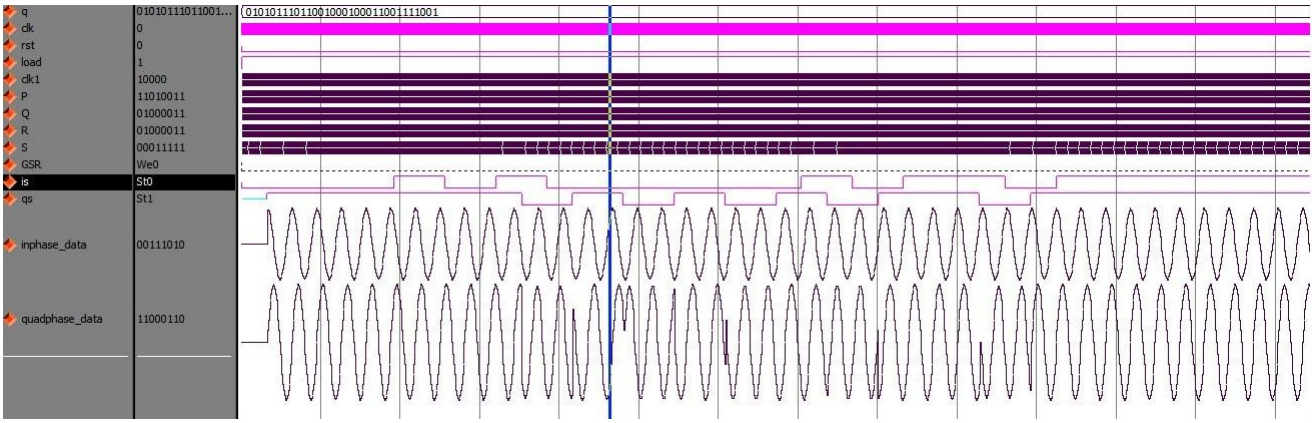

Fig. 8. Simulated output of QPSK

III. CONCLUSION

All the above mentioned modulation techniques have been simulated successfully. Proposed method of implementation proved to be very resilient and efficient [11]. The architecture used for the simulation purpose is mainly combinational and therefore do not contribute much propagation delay during simulation. The discussion and tabulation on power consumption, propagation delay(speed) and different area parameters i.e. piece of slice flip-flops, figures of 4 input LUTs, statistics of occupied slices, total number of 4 input LUTs used, number of bonded IOBs etc. have been accomplished. This whole simulation is tested for clock of 50 MHz frequency [12].

Future works includes simulation of more advance modulation techniques such as Offset Quadrature phase shift keying(OQPSK), Quadrature amplitude modulation(QAM), Minimum shift keying(MSK), Continuous phase frequency shift keying(CPFSK) and more.

IV. ACKNOWLEDGMENT

This work has been carried out under SMDP-C2SD project funded by DeitY, Govt. of India in the Department of Electronics and Telecommunication Engineering at National Institute of Technology, Raipur, India. Authors are thankful to the Ministry for the facilities provided under this project.

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